

830,000 m² of green land built over?

*Testing a newsroom's satellite estimate of land lost to development in Malta
against its own map, an independent land-cover model and EEA data*

THE CLAIM UNDER REVIEW

***“Nearly 830,000 square metres of nature
and cropland have been built up in Malta
between 2018 and 2023.”***

Amphora Media, Green to Grey, 11 September 2026.

Companion piece: nearly 95% of the take-up was farmland.

VERDICT: LARGELY SUPPORTED

Matches Amphora's own map; not independently confirmed

TL;DR

In September 2026 Amphora Media, an investigative newsroom, reported that “nearly 830,000 square metres of nature and cropland have been built up in Malta between 2018 and 2023”, and that in “nearly 95% of the take-up, farmland rather than natural area was affected”. The press is held to the same standard as everyone else here. We checked the arithmetic, the map Amphora published, an independent satellite land-cover product and the EEA’s land-change maps.

- 1 Amphora’s own map adds up.** The 397 change polygons behind Amphora’s map cover 828,429 m², so “nearly 830,000” is exact; the headline’s “over” is not. 830,000 m² is 0.26% of Malta’s land and 116 football pitches, as stated.
- 2 An independent model does not confirm which land changed.** Impact Observatory’s 10 m maps already show 68% of Amphora’s area as built-up in 2018 and class only 6–14% of it as newly built; only 2–3% of the model’s own new built-up land lies inside Amphora’s polygons. Its national total ranges from about zero to 3 km² net, depending on the years compared. Version 1.1 read it as support; it measures different land and can neither confirm nor rule out 0.83 km².
- 3 The 95% can be reproduced, with two choices.** Of the area with a known 2018 class, 34% was cropland and 61% grass: 96% together. That counts Dynamic World’s “grass”, which by its definition includes natural meadows, as farmland, and leaves out the 37% of the area with no 2018 class. This is our reading; Amphora has not stated its basis.
- 4 The EEA figures check out.** The EEA’s CORINE change maps show 20.6 ha of new artificial land in 2006–2012 and 93.7 ha in 2012–2018, close to the 190,000 and 920,000 m² Amphora quotes. They map only changes of 5 ha or more.
- 5 Verdict: largely supported (moderate confidence).** The figure is Amphora’s own checked map and the sums are right. But no independent source confirms the land itself, the 95% depends on how grass and unknown land are counted, and the headline rounds up.

THIS CLAIM



SUPPORTED

LARGELY
SUPPORTEDNOT
SUBSTANTIATED

MISLEADING

CONTRADICTED

0.83

km² in Amphora’s 397 published polygons (828,429 m²)

2–3%

of an independent model’s new built-up land lies inside them

96%

cropland or grass, of the area with a known 2018 class (37% unknown)

0.26%

of Malta’s land, as stated

WHAT WOULD MOVE THE VERDICT UP

Imagery checks showing a sample of Amphora’s polygons was green land in 2018 and built by 2023, and Amphora’s statement of the basis of the 95%.

WHAT WOULD MOVE IT DOWN

Imagery showing many polygons were already built before 2018, as the independent model suggests, or that the 95% leaves out land that was not farmland.

IN THIS REPORT

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1 The claim and what we could verify

Amphora Media published the investigation on 11 September 2026 [1] with a companion piece on agricultural land the next day [2], in collaboration with Arena for Journalism in Europe and supported by The Malta Environment Foundation. We read both in full. The change polygons behind the article’s interactive map are a public file on Amphora’s website [8]; we downloaded it on 5 October 2026.

What was said	Where	Access
“nearly 830,000 square metres of nature and cropland have been built up in Malta between 2018 and 2023” (headline: “over 830,000”)	Amphora [1]	Read in full
Total land loss 0.26%, “among the top 5 countries”; “a conservative estimate”	Amphora [1]	Read in full
EEA: 920,000 m² (2012–2018) and 190,000 m² (2006–2012); “at least 1.94 square kilometres” since 2006	Amphora [1]	Checked [7, 9]
“in nearly 95% of the take-up, farmland rather than natural area was affected”	Amphora [2]	Read in full
The map: 397 change polygons, each with its 2018 class (Dynamic World) and its status (“Built / developed”)	Amphora [8]	Downloaded

2 Method

Questions. Does Amphora’s map add up to 830,000 m², does independent data confirm the land it flags, are the comparisons right, and was the land mostly farmland?

Evidence. Amphora’s method [1] uses Google and WRI’s Dynamic World [3], with each flagged area checked by eye. We measured its published polygons in UTM 33N [8]; the file has no stated licence, so we do not redistribute it and record its SHA-256 instead. For an independent check we used a different model, Impact Observatory and Esri’s 10 m annual land cover [4, 5]: 2017–2023 through Microsoft Planetary Computer and, new in version 1.2, 2024 and 2025 from Esri’s image service [10], whose 2017–2023 maps are identical to the first set pixel for pixel. We laid Amphora’s polygons over these maps (pixel centres, with a 20 m tolerance as a check). The EEA’s CORINE Land Cover change maps [9] test the EEA figures. Island areas are from OpenStreetMap. Scripts and data: [tools/cc-019-report/](#) and [data/cc-019/](#).

Grades. Satellite land cover is a direct, model-classified measurement (grade B); CORINE and the EEA chart are official statistics (grade C). ♦ marks a source known second-hand.

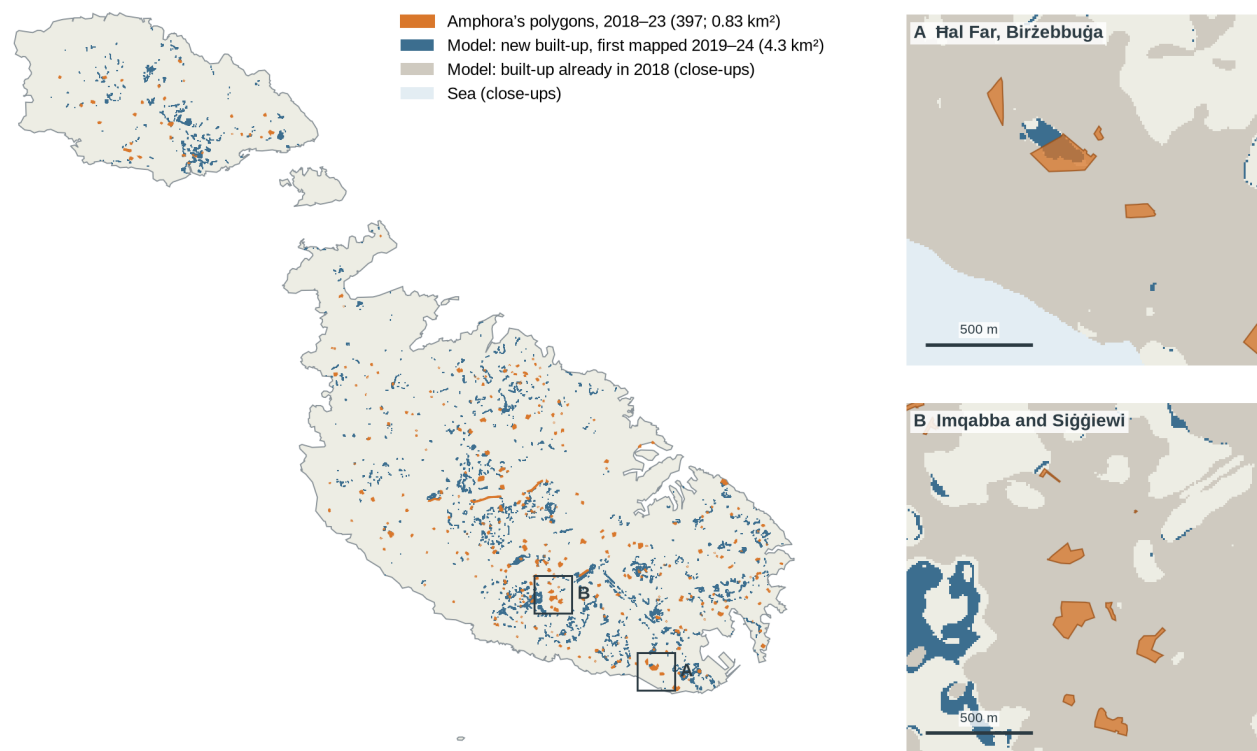
3 How the satellite data work

Both products classify every 10 m pixel with a deep-learning model trained on Sentinel-2 imagery [3, 4]. Single-year maps are noisy: in the Impact Observatory data Malta’s built-up area changes by 3 to 27 km² from one year to the next (–21 km² from 2023 to 2024, +23 km² from 2024 to 2025); the largest swing is over thirty times the 0.83 km² being measured. Comparing the 2018 and 2023 maps directly gives 24 km² of newly built-up pixels, or 18.9 km² net, which is not credible. We therefore counted only pixels that changed consistently: not built up in each of several early maps and built up in each of several later ones, and subtracted the reverse change as a gauge of noise. Each rule sees only land first mapped as built in a particular window: 2020–21 (3-year rule, maps 2017–19 against 2021–23), 2019–22 (2-year rule), and, with the 2024 and 2025 maps, 2020–23 and 2019–24. Amphora handled the same problem by checking each flagged area by eye, which removes false alarms but can miss developments, as it notes.

4

What the data show

Where Amphora and the independent model see new building



Only 2.7% of the model's new built-up pixels fall inside Amphora's polygons, and the model already mapped 68% of Amphora's area as built-up in 2018.

Sources: Amphora Media, Green to Grey change polygons (amphora.media, retrieved 5 Oct 2026); Impact Observatory / Esri 10 m annual land cover (2017–2025); coastline © OpenStreetMap contributors.

Figure 1. Amphora's polygons (orange) over the land the independent model maps as newly built-up, first mapped in 2019–24 (blue). Close-ups: grey is land the model already mapped as built-up in 2018. The two rarely coincide. Polygons: Amphora Media [8].



Left: net = new built-up minus the reverse change under the same rule; dotted lines are gross. Windows: when land was first mapped as built-up. Right: Amphora's 2018 classes come from Dynamic World (cropland, grass, other; 37% unknown); the model's classes are Impact Observatory's.

Figure 2. Left: Amphora's figure against the independent model's net totals under four rules; the result swings from 3.0 to below zero with the years used. Right: what the land was in 2018, in Amphora's own classes and in the model's.

Check	Result	Amphora	Grade
Area of Amphora's 397 polygons [8]	828,429 m²	nearly (headline: over) 830,000 m ²	–
830,000 m ² as a share of Malta's land (314 km ²)	0.264%	0.26%	C
In FIFA pitches; vs Comino; vs Manoel Island	116; 0.24–0.30; 2.7	116; roughly a quarter; two	C
Cropland + grass, of the area with a known 2018 class [8]	96% (95.3–95.7%)	nearly 95% farmland	–
Area with no 2018 class; polygons that were sea in 2018 [8]	37%; 2 (4,429 m ²)	–	–
Amphora's area already built-up in the 2018 map [5]	68% (61% in all of 2017–19)	–	B
Amphora's area the model maps as newly built [5]	6–14%	–	B
Model's new built-up land inside Amphora's polygons [5]	2–3% (5–7% within 20 m)	–	B
Model's net new built-up, four rules [5, 10]	–0.06 to 2.96 km ²	0.83 km ²	B
Model: new built-up was crops / rangeland / bare (3-year rule)	65% / 31% / 4%	nearly 95% farmland	B
CORINE: new artificial land 2006–12 / 2012–18 [9]	20.6 ha / 93.7 ha	19 ha / 92 ha	C
EEA chart, land take 2012–18 [7]	0.91–0.94 km ² ; highest of 39	0.92 km ²	C
EEA 2006–12 + 2012–18 + Amphora 2018–23	1.94 km ²	at least 1.94 km ²	C

All values in *data/cc-019/checks.csv*, *amphora_classes.csv* and *amphora_io_overlap.csv*. Comino: 0.30 of its 2.8 km² OpenStreetMap outline, 0.24 of the 3.5 km² often quoted. Amphora's figures have no grade: they are the claim being tested.

5

Where the evidence points different ways

Q1 Is 830,000 m² right?

PLAUSIBLE, UNCONFIRMED

EVIDENCE FOR THE CLAIM'S IMPRESSION

Amphora published its polygons, which add up to 828,429 m², and says each area was checked by eye. The model agrees that 88% of that area was built-up by 2023 and 92% by 2025.

EVIDENCE AGAINST

The model already maps 68% of the area as built-up in 2018, and only 2–3% of its own new built-up land lies inside Amphora's polygons. Its national total swings from about zero to 3 km² net with the years compared.

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: the two disagree mainly about what was there in 2018. Without checking imagery we cannot say which is right; the figure stands on Amphora's own checks, not on independent confirmation.

Q2 Was 95% of it farmland?

DEPENDS ON DEFINITIONS

EVIDENCE FOR THE CLAIM'S IMPRESSION

In Amphora's own data, cropland and grass make up 96% of the area with a known 2018 class. Amphora may count grassland as farmland; it has not said.

EVIDENCE AGAINST

Dynamic World's "grass" includes natural meadows and excludes plotted fields, which it calls crops [3]. 37% of the area has no 2018 class; cropland alone is 22% of the whole. The model puts 65% in crops and 31% in rangeland.

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: "nearly 95%" can be reproduced only by counting all grass as farmland and leaving out the unknown third.

Q3 Over or nearly 830,000?

HEADLINE ROUNDS UP

EVIDENCE FOR THE CLAIM'S IMPRESSION

The text says "nearly", and the polygons add up to 828,429 m².

EVIDENCE AGAINST

The headline says "over" 830,000.

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: a minor point that does not change the finding.

6

Testing the claim

Sub-claim	What the evidence shows	Rating
A. About 830,000 m² of green land built up 2018–2023	Amphora's polygons add up to 828,429 m² (two, 4,429 m², were sea in 2018). An independent model flags largely different land and can neither confirm nor rule out the figure.	PLAUSIBLE
B. 0.26% of land; comparisons (pitches, Comino, Manoel)	Arithmetic checks out (Comino: 0.24 of the 3.5 km² often quoted, 0.30 of the OSM outline; Manoel: 2.7, not two).	SUPPORTED
C. Nearly 95% of take-up was farmland	Cropland + grass = 96% of the area with a known 2018 class (our reading of the basis), but only if all grass is farmland; 37% of the area has no class.	NEEDS CONTEXT
D. EEA figures and the 1.94 km² total	CORINE change maps: 20.6 ha (2006–12) and 93.7 ha (2012–18) [9]; EEA chart 0.91–0.94 km² [7]; sum correct.	SUPPORTED

7 Verdict and requests for evidence

VERDICT

LARGELY SUPPORTED

The figure matches Amphora's own published map; no independent source confirms the land, and the 95% depends on definitions. Confidence: moderate.

Why. (1) The figure is the area of Amphora's published polygons, which it says it checked by eye, and the arithmetic, comparisons and EEA figures are right. (2) An independent model neither confirms nor contradicts it: the two flag largely different land, and the model's totals swing with the years compared. Version 1.1 read the model as support; that went too far. (3) The 95% farmland share can be reproduced only by counting grass as farmland and leaving out land of unknown class, and the headline rounds up a number the text rounds down. These are caveats on how far the figure has been confirmed, not evidence that it is wrong.

Evidence we are asking for

- ☐ Amphora: the basis and period of the 95% farmland figure, and how "grass" was counted.
- ☐ Amphora: whether the polygons were checked against 2023 or 2025 imagery (the map file says 2018–2025), and the 2018 class of the 162 polygons marked unknown.
- ☐ Anyone: a check of a sample of polygons against high-resolution imagery from 2018.

8 Limitations

- Land-cover models misclassify pixels; our rules reduce but do not remove this. We did not inspect imagery, so we cannot say whether Amphora or the model is right about what stood on these parcels in 2018.
- Amphora's polygons are small (median 504 m², about five pixels); a 20 m tolerance raises the overlap only to 5–7%, so misalignment does not explain it.
- The persistence rules only see land first mapped as built in set windows (2019–24 at most), not exactly 2018–2023.
- Amphora's map file gives its years as 2018–2025 and each polygon's status as of 2025; the article says 2018–2023.
- CORINE maps only changes of 5 ha or more; every one of Amphora's polygons (the largest is 3.3 ha) is below that.
- Built-up land includes roads, car parks and quarries as well as buildings; so does Amphora's. Neither estimate says whether development was inside or outside the development zone.
- Class names differ between the two models; 'rangeland' in IO is not the same as 'natural'.

References

- [1] Amphora Media (11 Sep 2026). Green To Grey: Malta lost over 830,000 square metres of land to development in five years. <https://www.amphora.media/2026/09/green-to-grey-malta-land-loss-development-construction>
- [2] Amphora Media (12 Sep 2026). Green to Grey: when development creeps up on agricultural land. <https://www.amphora.media/2026/09/green-to-grey-development-agricultural-land>
- [3] Brown C.F. et al. (2022). Dynamic World, near real-time global 10 m land use land cover mapping. *Scientific Data* 9:251. doi:10.1038/s41597-022-01307-4. (Class definitions, Table 1, read.) <https://doi.org/10.1038/s41597-022-01307-4>
- [4] Karra K. et al. (2021). Global land use / land cover with Sentinel 2 and deep learning. *IGARSS 2021*, 4704–4707. doi:10.1109/IGARSS47720.2021.9553499. <https://doi.org/10.1109/IGARSS47720.2021.9553499>
- [5] Impact Observatory and Esri. 10 m Annual Land Use Land Cover (9-class) V2, 2017–2023; Microsoft Planetary Computer; retrieved 4 Oct 2026. <https://planetarycomputer.microsoft.com/dataset/io-lulc-annual-v02>

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- [6] MiŻien. Data and calculations: [data/cc-019/](#); [tools/cc-019-report/](#).
- [7] European Environment Agency (2019, modified 2024). Country comparison: land take and land recultivation in EEA39 in the period 2012–2018 (in proportion of country area). Value read from the chart. <https://www.eea.europa.eu/en/analysis/maps-and-charts/country-comparison-land-take-and>
- [8] Amphora Media (2026). Green to Grey change polygons for Malta, [malta-change.geojson](#) (generated 30 Aug 2026; retrieved 5 Oct 2026; SHA-256 cf250ece...fcf320). Not redistributed. <https://www.amphora.media/greentogrey/data/malta-change.geojson>
- [9] European Environment Agency. CORINE Land Cover change layers 2006–2012 and 2012–2018 (CHA0612, CHA1218), [discomap](#) map service; queried 5 Oct 2026. <https://image.discomap.eea.europa.eu/arcgis/rest/services/Corine>
- [10] Impact Observatory, Microsoft and Esri. Sentinel-2 10m Land Cover time series, 2017–2025 (Esri Living Atlas image service); retrieved 5 Oct 2026. https://ic.imagery1.arcgis.com/arcgis/rest/services/Sentinel2_10m_LandCover/ImageServer

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient, or direct measurement · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	4 Oct 2026	First issue. Right of reply to Amphora Media not yet sent.
1.1	5 Oct 2026	Corrections. (1) Periods of the independent estimates: “2018–2023” → land first mapped as built in 2020–21 (3-year rule) and 2019–22 (2-year rule), which is what the rules can detect (TL;DR, tile, sections 3, 4, 5 and 6, limitations, Figure 2 labels and caption). (2) Comino: “about 0.3 of Comino, as stated” → Amphora says “roughly a quarter”; 830,000 m² is 0.24 of the 3.5 km² often quoted and 0.30 of the 2.8 km² OpenStreetMap outline. (3) Simple comparison: “would give 24 km²” → 24 km² of newly built-up pixels, 18.9 km² net (now saved in <i>data/cc-019/io_lulc_change.csv</i>). (4) Year-to-year noise: “more than 15 km², twenty times the change” → 3 to 27 km², the largest over thirty times 0.83 km²; flyer “15 km² a year” → “up to 27 km² from year to year”. (5) The TL;DR now says the two estimates are not like for like. Verdict and confidence unchanged.
1.2	5 Oct 2026	Upgrade. (1) Amphora’s published change polygons [8] measured: 397 polygons, 828,429 m², so “nearly 830,000” is reproduced and the headline’s “over” is not. (2) Spatial cross-check with the independent model: “an independent measurement finds more, not less” and “plausible and conservative” → the model already maps 68% of Amphora’s area as built-up in 2018, 6–14% as new, and only 2–3% of its new built-up land lies inside Amphora’s polygons, so it neither confirms nor rules out the figure (TL;DR, tiles, sections 2–8, flyer). (3) The 2024 and 2025 maps added from Esri’s service (identical to the earlier maps for 2017–2023): net totals –0.06 to 2.96 km² by rule. (4) Sub-claim A “supported” → “plausible”; sub-claim C “not shown” → “needs context”: cropland + grass = 96% of the area with a known 2018 class (our reading of the basis of “nearly 95%”). (5) Sub-claim D: the 2006–2012 EEA figure checked with the CORINE change map (20.6 ha; 93.7 ha for 2012–2018). (6) New Figure 1 (Amphora’s polygons over the model’s new built-up land), replacing the map of new built-up pixels; Figure 2 redrawn with four rules and Amphora’s own 2018 classes. Verdict and confidence unchanged.